

Machine Learning Project

Three datasets · classical supervised learning · scikit-learn

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1. Which supervised models perform best for each dataset?
2. Do tuned models consistently outperform baseline models?
3. How do classification and regression results differ across domains?
4. How do our results compare with prior published work?

DATASET 1

Spotify Yearly Track Metadata

12,397

records

**5-class query
+ duration
regression**

Metadata only: year, month, day, artists,
name length

Selected for dual-task workflow

DATASET 2

Heart Disease UCI/Kaggle

1,025

records

**Binary disease
classification**

Clinical features: chest pain, ST
depression, exercise angina

Selected for medical relevance

DATASET 3

Housing Prices Kaggle

545

records

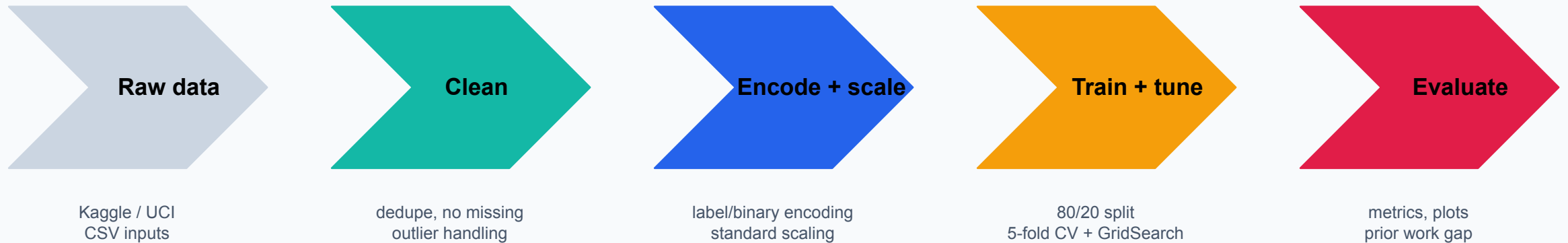
**Continuous
price
regression**

Numeric + binary property attributes; no
missing values

Selected as pure regression case

Modeling Workflow

CSCI 164 Final Project



Preprocessing choices that mattered

- Spotify: deduplicated by track_id; removed >15 minute outliers.
- Housing: binary/ordinal encoding; target kept in rupees.

Validation discipline

- Classification splits were stratified to preserve class balance.
- Cross-validation compared models before test-set claims.

Model development and tuning

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Dataset	Models used	Best tuned setting	Main lesson
Spotify clf	LogReg · Tree · k-NN	k=31, Manhattan, distance	Metadata can separate “remix”; weak for mood/search intent
Spotify reg	Linear · Tree regressor	Tree depth=5, split=20	Duration is weakly predicted from metadata
Heart	LogReg · Tree · k-NN · NB	k=21, Manhattan, distance	Local patient similarity worked best
Housing	Linear · Ridge · Tree · k-NN	Ridge alpha=50, k-NN k=9	Linear structure beat trees

GridSearchCV

5-fold search on train set for tunable models

Shallow trees

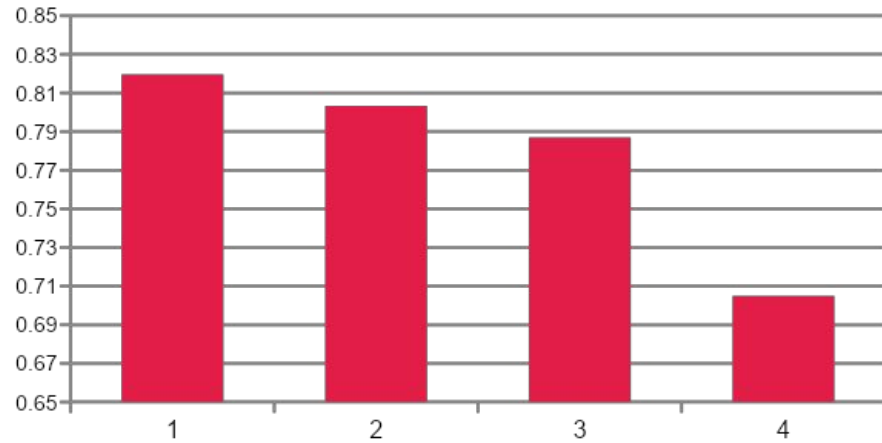
max_depth≈5 generally beat deeper trees

Large k values

k=21–31 reduced variance in sparse tabular settings

Classification results

Heart Disease: test performance



81.97%

best accuracy

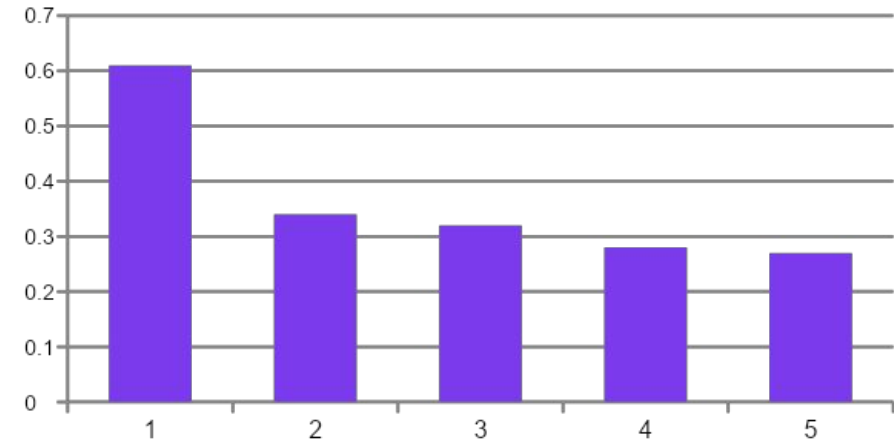
0.895

ROC-AUC

85%

disease recall

Spotify: class-level weakness



37.3%

best accuracy

20%

random baseline

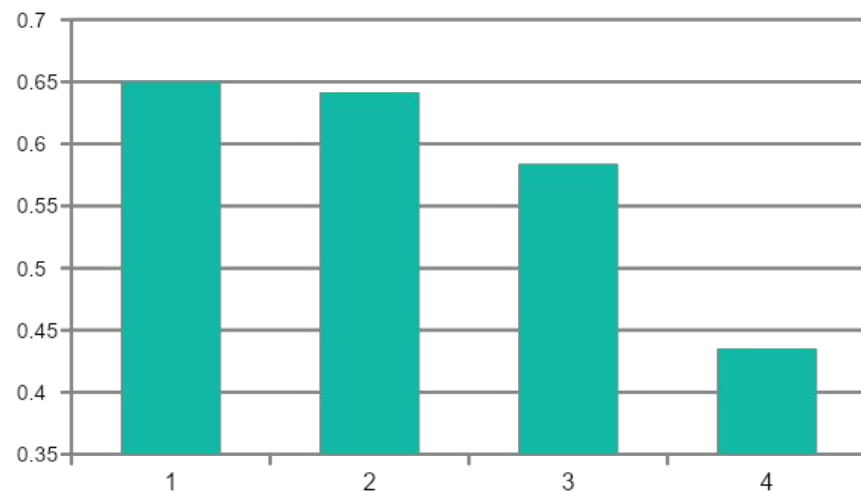
0.61

remix F1

The takeaway is heart disease performed well because clinical features were strongly predictive. Spotify was harder because metadata alone did not capture enough musical information.

Regression results

Housing price: R^2 by model



0.650

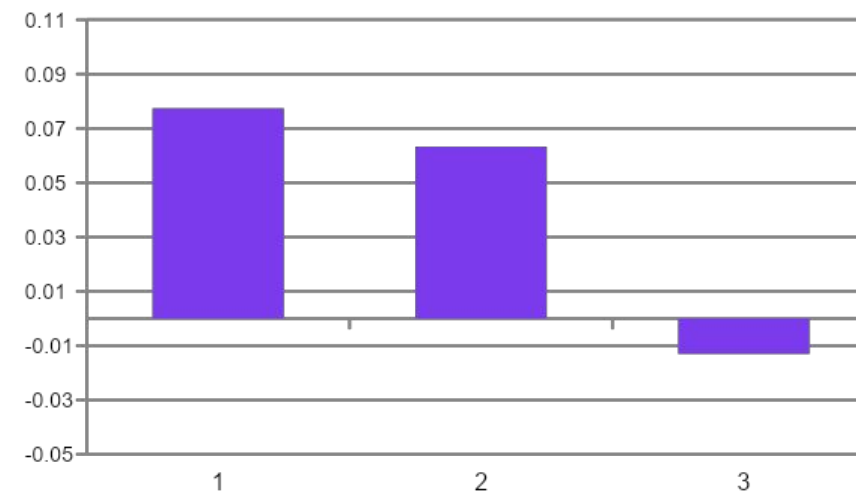
best R^2

0.98M

best MAE

Takeaway: small tabular housing data favored interpretable linear structure.

Spotify duration: R^2 by model



0.077

best R^2

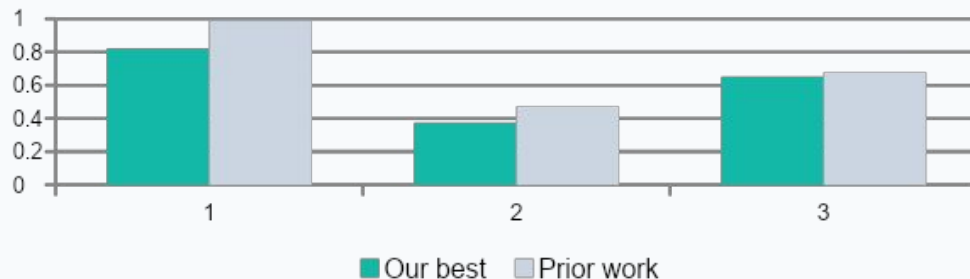
0.768 min

best MAE

Takeaway: metadata only weakly explains track duration.

Literature comparison

Dataset	Prior benchmark	Our best result	Gap diagnosis
Heart Disease	99% accuracy RF/deep learning pipeline	81.97% accuracy AUC 0.895	Classical shallow models vs. more complex ensemble/deep approach
Spotify	47.27% LR accuracy audio features + SMOTE	37.26% accuracy metadata-only k-NN	No audio features; labels are search-query categories
Housing	Linear $R^2 \approx 0.678$ similar regression study	Linear $R^2 = 0.6495$	Close match; different rows/features and high-price variance



Critical read

Our results were generally lower than prior published work, mainly because our models used simpler scikit-learn methods and more limited feature sets. The gaps were largest for Spotify and Heart Disease, while Housing closely matched the prior regression benchmark.

What we learned

Finding 1

k-NN was the strongest classifier

Local similarity worked for Heart Disease and marginally for Spotify.

Finding 2

Linear models dominated housing

The property-price relationship was more linear than tree-like in this small dataset.

Finding 3

Feature quality beat algorithm complexity

Spotify underperformed because metadata could not encode audio/mood.

Limitations that should be stated plainly

- Small Heart and Housing datasets raise variance risk.
- No Random Forest, Gradient Boosting, or deep models in final scope.
- Spotify lacked audio features; Housing lacked geography.

Next improvements with highest expected return

- Add richer features: audio attributes, location, interaction terms.
- Try ensembles after baselines: Random Forest, XGBoost-style GBM.
- Log-transform housing price to reduce high-end residual spread.

Final takeaway: The best model depended on the dataset k-NN worked best for classification, linear regression worked best for housing, and feature quality was the biggest factor limiting performance.

Thank You